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Research on Path Planning Optimization Based on Genetic Algorithm

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Abstract—In real life, robots are subject to multiple constraints when moving in complex scenes, which can lead to problems such as poor optimization ability, unsmooth path planning, and slow convergence speed of search algorithms. A robot path planning method based on improved genetic algorithm is proposed to address the above issues. Firstly, a grid method is used to construct the mobile environment of the robot, and a model is constructed under constraints such as path smoothness, path length, and path difficulty; Then, the traditional genetic algorithm was improved through smoothing operators and its performance was verified through experiments. The simulation results show that the algorithm proposed in this paper can effectively handle path planning problems under multiple constraint conditions and find the most suitable smooth path for robot motion. Through comparative experiments, this method has relative advantages in path length, smoothness, and runtime.

Keywords: robot, genetic algorithm, path planning, grid method

I. Introduction

Currently, intelligent robots are very widely used in several fields of production life, such as industry, storage and logistics, medicine and agriculture, and path planning is one of the hot technologies in the research field [1-2]. To study path planning is to be able to enable a robot to find an optimal or collision-free obstacle-free path between a set starting point and an end point according to a special scenario, specific criteria, and an environment full of obstacles [3]. In the path planning problem, the efficiency of robot path planning, the time taken to compute the path, the complexity of the environment, the static or dynamic scenario, etc. are all factors that influence [4-5]. Therefore, it is a very challenging problem to find an optimal smooth path that avoids obstacle interference and at the same time can be found.

To tackle these challenges, nature-inspired approaches are being adopted nowadays due to their capability to explore the global optimum path in complex environments [6]. In early research, the best solution for path planning was found mainly by evolutionary algorithms [7]. However, evolutionary algorithms require the location of the grid to be pre-fixed when modeling the environment, which can make the flexibility of path planning limited by the size of the grid [8].

To solve the above problems, many experts and scholars have conducted a lot of research on this. Pere [9] et al. proposed a robot path planning algorithm with improved fast extended random tree, which can effectively solve the path planning problem of robots in complex environments by establishing a search tree model to improve the robot path perception. Qu et al [10] proposed a hybrid gray wolf path planning optimization algorithm to speed up the convergence speed and preserve the population exploration capability by simplifying the computation stage, so as to obtain effective optimal movement path, but the path planning efficiency of the above method is relatively low and easy to fall into local minima.

To address the above problems, this paper proposes a robot path planning method based on improved genetic algorithm. The path length, path smoothness, path difficulty and other constraints are used to construct the environment model by raster method, and the SPS (Surrounding Point Sel) method is used to solve the path generation problem near the obstacles. At the same time, delete operator and smoothing operator are introduced to improve the traditional genetic algorithm to improve the smoothness of paths and increase the convergence speed of the algorithm.

II. Constructing the environment model

The path planning is to find a collision-free path from the initial point to the target one. The grid-based algorithms play an important role in path planning. Many path planning methods use a grid-based model to represent the environment space as shown in Fig. 1. It has been determined that calculation of distance and representation of obstacle is easier with grid-based representation. The grid-based environment space is represented in two ways, by the way of an orderly numbered grid [1,13–15] or by the way of (x, y) coordinates plane [9,16]. It has been found that in the literature the coordinates plane representation is widely used, therefore this representation method is used in the present study.

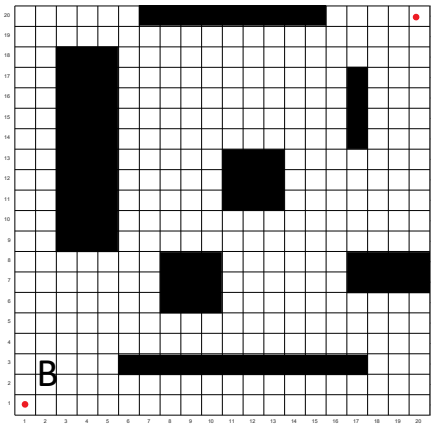


Figure 1 Grid-based environment representation

The feasible path for one of the robots is as in equation (1):

$$p = \{p_1, p_2, p_3, \dots \dots p_n\},$$
$$p_1 = B, p_n = E, p_j \cap Obs_j = \emptyset \quad (1)$$
$$1 \le i \le n. 1 \le j \le n$$

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Where, n is the number of nodes contained in the set p of robot walking paths; pi is a certain i-node on the path, and the robot's walking path under the genetic algorithm is .

$$P^* = \{P_1^*, P_2^*, P_3^* \ldots \ldots P_m^*\}, P^* \in P \quad (2)$$

III. Description of bidirectional smoothing genetic algorithm

In order to address the problems of poor robot path planning seeking ability, unsmoothed paths and low search efficiency under multiple constraints, this paper optimizes and improves the traditional genetic algorithm for the problem.

A. Individual coding

In this paper, the working environment of the robot is a uniformly distributed two-dimensional space constructed by grid points, and each grid point is represented by coordinates (xi，yi), where xi is the horizontal coordinate of the path node, yi is the vertical coordinate of the path node, and the line formed by different nodes is the driving path of the robot, and the sum of the path lengths between nodes is the path length of the robot.

The problem of random initialization is that after initialization, it is necessary to consider whether these randomly generated nodes are useful for path planning, and when there are too many useless nodes, it will certainly lead to an increase in the computational cost of the algorithm and even affect the acquisition of the optimal path. In order to solve the above-mentioned problems in the random initialization of the population, the traditional genetic algorithm population initialization is improved with the following steps:

Step 1: In the two-dimensional space where the robot works, the starting point and the target point of the robot are first determined. where the position of the robot is located at the starting point;

Step 2: There are no obstacles at the starting and target points, and they move from the starting point to the target position in the direction of the shortest straight distance. If there are obstacles along the way, a point set is generated around the obstacles using the SPS algorithm, as shown in Figure 2, which is a schematic diagram of the process:

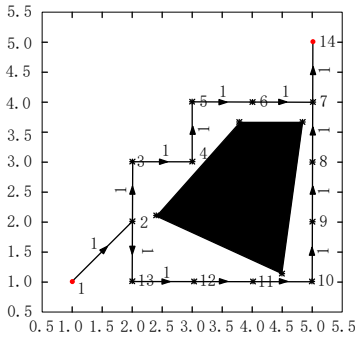


Figure 2 Point set generation around the obstacle by SPS algorithm

Step 3: Test the feasibility of the path points generated by SPS algorithm by Dijkstra algorithm, if it is feasible then continue walking, if the path points are not feasible then reduce the width of the grid points, and then return to step 2.

Step 4: Determine whether the robot coordinates have reached the target location, if they have, then save the current path from the starting point to the target point, if it has not reached the target location, then return to step 2.

Step 5: Calculate the number of individuals in the population, if the population is greater than or equal to 2M, then end the iteration; if it is less than 2, then repeat the above steps from step 1.

B. Adaptation value evaluation and assignment

1) Fitness function

Considering the complexity of robot path planning, this paper determines the fitness function based on three objectives: path difficulty, smoothness and path length.

2) Adaptation value assignment

The adaptation value can evaluate the quality of the feasible solution during the iteration of the algorithm. In this paper, we use the SPEA2 method to assign adaptation values to individuals in the population, and the specific assignment method is as follows:

Step 1: Initialize the population to obtain a population P of size N, create another empty set of size N′ , and then replicate the population P into P′ ;

Step 2: Delete the individuals in P′ that are dominated by other individuals, if the number of individuals in P′ is > N′, then the set P′ needs to be processed by a clustering algorithm to reduce the number of individuals in it;

Step 3: The individuals in P′ can be solved for adaptation values according to equation (3).

$$S(i) = \frac{t}{N+1} \quad (3)$$

where t is the number of individuals in P that are at the disposal of i in the set P′.

Step 4: Assign the adaptation values shown in Eq. (4) to the individuals in the population P

$$f(i) = 1 + \sum S(j) \quad (4)$$

Where, j the set P' of individuals that can dominate individual i; S(j) is the adaptation value of individual j in the set P′.

In this paper, by duplicating P to P' duplicate solutions for judgment, if there are duplicate solutions, then delete them, which can avoid the problem of early convergence of individuals with high adaptation value due to over reproduction caused by duplicate solutions.

C. Selection operator

In this paper, we use a combination of roulette selection strategy and elite selection strategy to design the selection operator, and the schematic diagram is shown in Figure 3. It can ensure the randomness of the selection operation while avoiding the loss of the optimal individuals, thus improving the convergence speed of the algorithm.

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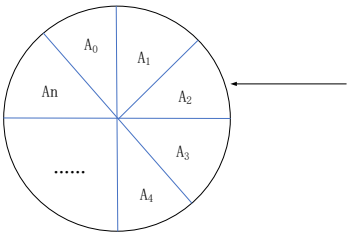


Figure 3 Diagram of roulette selection strategy

D. Crossover operator

Randomly select two individuals and cross the same path points of two random individuals, if the same path points are greater than or equal to 2, then select any one of the same path points for single point crossover; if there is no same path point, then select one path point from each of the two individuals, that is, cross the path points selected on these two individuals, if the path formed after crossover is a discontinuous path, then by SPS algorithm The disconnected paths are reconnected to form continuous paths. The schematic diagram of the genetic algorithm crossover operator is shown in Figure 4.

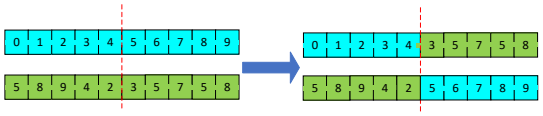


Figure 4 Single-point crossover diagram

E. Mutation operators

In nature, a small number of newborn offspring may have a small probability of mutations in some of their genes, resulting in the appearance of traits different from those of their "parents". Mutations take care of the diversity among populations and prevent premature convergence. In practice, a small probability of mutation is usually set, and if the probability of mutation is large, the whole search process degenerates into a random search process. Therefore, it is more prudent to take a relatively large value of probability p at the beginning of the evolutionary process, and gradually reduce p to around 0 as the search process proceeds. The schematic diagram of the genetic algorithm mutation operator is shown in Figure 5.

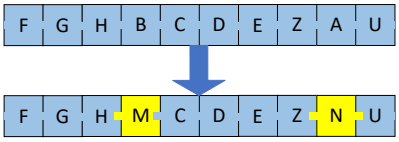


Figure 5 Schematic diagram of offspring variation

F. Improved genetic algorithm smoothing operator

In practice, mobile robots are not fully suitable for the folded paths planned by the genetic algorithm, and some physical characteristics constraints need to be satisfied during the robot's travel. In this paper, Bézier curve (Bézier curve) is used for constraint optimization. Bézier curve was proposed by French engineer Pierre Bézier in 1962, which can satisfy the kinematic constraints of the vehicle and the fixed-point constraints of the obstacles, and is a more ideal path optimization method. n-order Bézier curve simplifies the solution of trajectory planning, and its control equation is shown in equation (5) is shown as follows:

$$P(t) = \sum_{i=0}^{n} \binom{n}{i} P_i (1-t)^{n-i} t^i \tag{5}$$

where P represents the fixed point, N is the number of nodes, t ∈ [0,1], and the nth-order Bessel curve is actually an equivalent interpolation of n-1 order. In this paper, we take the third-order Bessel curve as an example, and the curvature of the curve at the starting point can be calculated by equation (6).

$$k(0) = \frac{3}{4} \times \frac{|(P_1 - P_0) \times (P_2 - P_1)|}{|P_1 - P_0|^3} \tag{6}$$

In the path planned by the genetic algorithm, the intersection angle of its neighboring 3 nodes is calculated and the corresponding curve is optimized so that it bypasses the obstacle point and shortens the distance, the curvature relationship of the Bessel curve at a node can be expressed as

$$k(\pi) = \frac{x'(\tau)y''(\tau) - y'(\tau)x''(\tau)}{(x'^2(\tau) + y'^2(\tau))^{\frac{3}{2}}} \tag{7}$$

In this paper, the control of the node is chosen as the presence point of the obstacle i.e. [P0, P3], when τ=0.25 the trajectory optimization schematic is shown in Fig. 6.

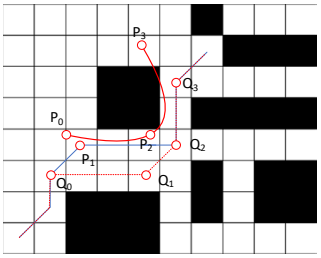


Figure 6 Principle diagram of trajectory optimization

The flowchart of the improved genetic algorithm proposed in this paper is shown in Fig. 7.

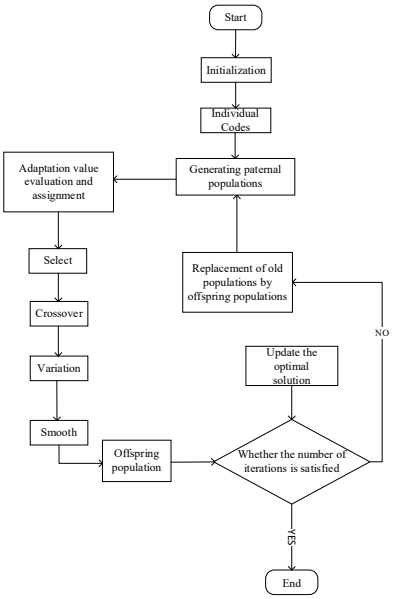


Figure 7 Flow chart of improved genetic algorithm

IV. Experiments and Analysis

In order to evaluate the performance of the proposed improved genetic algorithm, algorithm validity verification

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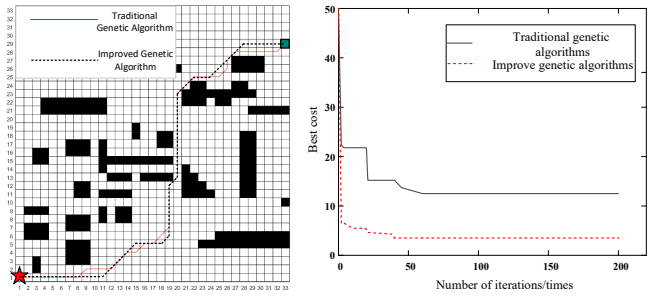
experiments and comparative analysis experiments were conducted, both of which were implemented on the MATALAB simulation platform. The parameter settings of the improved genetic algorithm in the experiment are shown in Table 1:

Table 1 Parameter design

Parameter design	Value
Population size	100
Maximum number of iterations	200
Crossover probability	0.8
Mutation probability	0.2
Path length weighting	1
Path smoothness weighting	7

A. Validity verification experiment

The path planning results of the proposed algorithm and the traditional genetic algorithm are shown in Figure 8, where the red pentagram is the starting point and the gray square is the destination. The convergence curves of the two algorithms are shown in Figure 9.



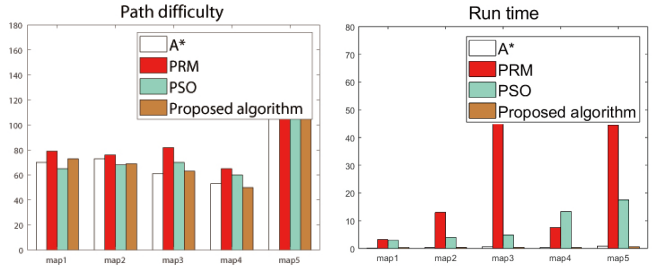
(a) (b)

Figure 8 (a): Comparison of two path planning algorithms (b): Convergence curves of two algorithms

From Figures 8, it can be seen that the improved genetic algorithm outperforms traditional genetic algorithms in terms of path length, path smoothness, and path difficulty.

B. Comparison experiments

The A*, PRM and PSO algorithms are selected for comparison and analysis with the proposed improved genetic algorithm. As shown in Figure 10, the average values of path length, smoothness, running time and path difficulty for 50 iterations executed by A*, PRM and PSO algorithms and the proposed improved genetic algorithm are shown. Figure 9 shows the path planning results of the selected comparative algorithms in five different map environments.



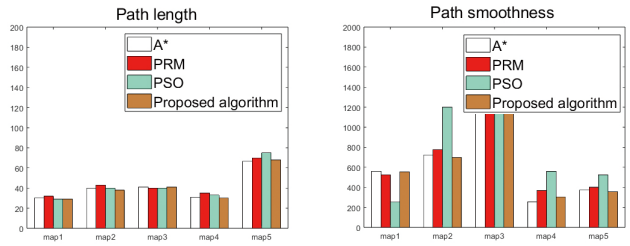


Figure 9 Comparison results of A*, PRM, PSO, and the proposed algorithm in terms of path length, smoothness, running time, and path difficulty

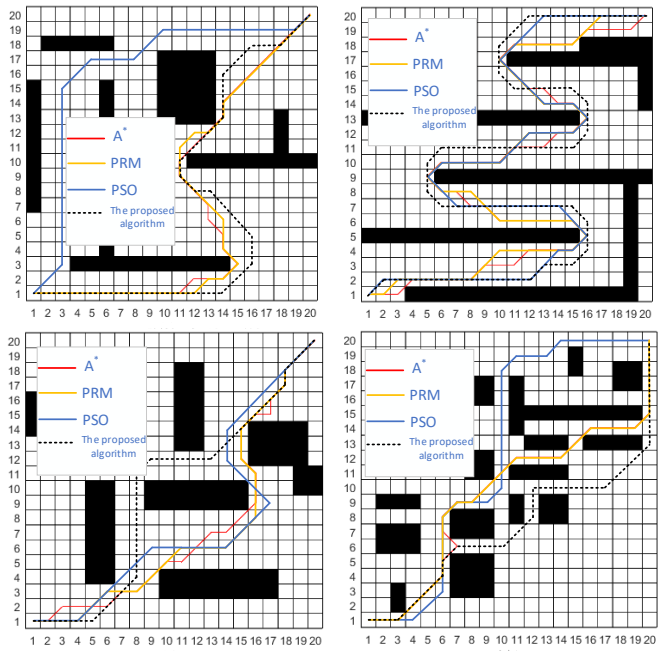


Figure 10 Path planning results under 4 different map environments

From Fig. 9 and Fig. 10, it can be seen that the shortest path is obtained by the proposed method in this paper in five map environments. Therefore, the proposed improved genetic algorithm shows the best performance in all environments.

V. Conclusion

To solve the problems of poor path finding ability and slow convergence of the search algorithm for robots under multiple constraints, a multi-robot path planning method with improved genetic algorithm is proposed, and the following conclusions are drawn through experimental analysis:

(1) The proposed method can effectively improve the iteration efficiency of the algorithm and obtain a smooth search path by adding a deletion operator and a smoothing operator to the objective function.

(2) Compared with other comparative methods, the proposed method can show the best performance in all environments with multiple objective functions considering path length, path smoothness, running time and path difficulty at the same time.

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References

[1] Johnson L B，Choi H，How J P’et al. The role of information assumptions in decentralized task allocation: A tutorial [J].IEEE Control Systems Magazine，2016，36(4):45-58. DOI: 10.1109/MCS.2016.2558419

[2] W. Chen, C. Sun, H. Liu, J. Liu and Y. Tang, "Path planning scheme for spray painting robot with Bézier curves on complex curved surfaces," 2017 32nd Youth Academic Annual Conference of Chinese Association of Automation (YAC), Hefei, China, 2017, pp. 698-703, doi: 10.1109/YAC.2017.7967499.

[3] Shubham S, Nk S, Vibhav K S, et al. Multi robot path planning based on cuckoo search and with PSO algorithm in the complex and the unknown environment[J]. Carbohydrate Research,2019,8(4):2924 -2932. DOI:10.35940/ijrte.D6813.118419

[4] Bonabeau E，Dorigo M，Theraulaz G. Swarm intelligence: from natural to artificial systems [M] . UK: Oxford University Press， 1999. https://doi.org/10.1093/oso /9780195131581.001.0001

[5] Wu Z S，Fu W P. A review of path planning method for mobile robot[J]. Advaced Materials Research ， 2014,1030-1032:1588-1591.https://doi.org/10.4028/www.scientific.net/AMR.1030-1032.1588

[6] A. Sonny, S. R. Yeduri and L. R. Cenkeramaddi, "Autonomous UAV Path Planning Using Modified PSO for UAV-Assisted Wireless Networks," in IEEE Access, vol. 11, pp. 70353-70367, 2023. DOI: 10.1109/ACCESS.2023.3293203

[7] Mahulea C, Montijano E, Kloetzer M. Distributed multirobot path planning in unknown maps using petri net mode[J].IFAC-Papers OnLine , 2020,53(2):2063 -2068. https://doi.org/10.1016/j.ifa col.2020.12.2521

[8] Nazarahari M,Khanmirza E, Doostie S.Multi-objective multi-robot path planning in continuous environment using an enhanced genetic algorithm[J].Expert Systems with Applications,2019,115:106-120. https://doi.org/10.1016/j.eswa.2018.08.008

[9] Pere. Automatic planning of manipulator movements[J].IEEE Trans on Sys Man andCyb,1981;11(11) DOI: 10.1109/TSMC.1981.4308589

[10] Chengzhi Qu, Wendong Gai, Jing Zhang, Maiying Zhong,A novel hybrid grey wolf optimizer algorithm for unmanned aerial vehicle (UAV) path planning, Knowledge-Based Systems, [J]. Knowledge-Based Systems ， 2020，194(4):1030 -1035. https://doi.org/10.1016/j.knosys.2020.105530.

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